12TH CHEMISTRY · CHAPTER 21 · FULL CHAPTER MCQS

Halogenoalkanes

94 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION AND CLASSIFICATION

1. CFCs were used in the past as:

(a) Fertilizers (b) Antibiotics
(c) Refrigerants (d) Fuels

Answer: (c) Refrigerants — Also as aerosol propellants.

2. PVC stands for:

(a) Polyvinyl carbonate (b) Polyvinyl cyanide
(c) Polyvalent chloride (d) Polyvinyl chloride

Answer: (d) Polyvinyl chloride — Used for piping.

3. PTFE (Teflon) is used for:

(a) Piping (b) Non-stick cookware
(c) Anesthesia (d) Refrigeration

Answer: (b) Non-stick cookware — Introduction.

4. Halothane is used as an:

(a) Inhalation anesthetic (b) Antiseptic
(c) Insecticide (d) Antacid

Answer: (a) Inhalation anesthetic — It causes reversible loss of consciousness.

5. Chloroform is no longer used as an anesthetic because of its:

(a) High cost (b) Toxicity
(c) Colour (d) Smell

Answer: (b) Toxicity — Safety concerns.

6. The halogens present in halothane are:

(a) F and Cl only (b) Cl and Br only
(c) F, Cl and Br (d) F and I

Answer: (c) F, Cl and Br — CF_3–CHClBr.

7. A halogenoalkane is formed when hydrogen of an alkane is replaced by:

(a) Halogen (b) Oxygen
(c) Nitrogen (d) A metal

Answer: (a) Halogen — Definition.

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8. Monohalogenoalkanes are also called:

(a) Aryl halides (b) Vinyl halides
(c) Acid halides (d) Alkyl halides

Answer: (d) Alkyl halides — General formula R–X.

9. The general formula of alkyl halides is:

(a) R–OH (b) R–O–R
(c) R–X (d) Ar–X

Answer: (c) R–X — X = F, Cl, Br, I.

10. In a primary halogenoalkane the halogen-bearing carbon is attached to:

(a) One or no carbon (b) Two carbons
(c) Three carbons (d) Four carbons

Answer: (a) One or no carbon — Definition.

11. 2-Chloropropane is a:

(a) Primary halogenoalkane (b) Secondary halogenoalkane
(c) Tertiary halogenoalkane (d) Dihalogenoalkane

Answer: (b) Secondary halogenoalkane — Two carbons on the C–Cl carbon.

12. 2-Chloro-2-methylpropane is a:

(a) Primary halogenoalkane (b) Halogenoarene
(c) Secondary halogenoalkane (d) Tertiary halogenoalkane

Answer: (d) Tertiary halogenoalkane — Three carbons on the C–Cl carbon.

13. Chloromethane is classified as:

(a) Secondary (b) Tertiary
(c) Primary (d) Quaternary

Answer: (c) Primary — No other carbon attached.

14. 2-Chlorobutane is a:

(a) Secondary halogenoalkane (b) Primary halogenoalkane
(c) Tertiary halogenoalkane (d) Trihalogenoalkane

Answer: (a) Secondary halogenoalkane — Section introduction.

15. The number of structural isomers of $C_5H_{11}Cl$ is:

(a) Four (b) Eight
(c) Six (d) Five

Answer: (b) Eight — Quick Check 21.1.

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16. The IUPAC name of $(CH_3)_2CH\text{-}CH_2\text{-}Cl$ is:

(a) 2-Chlorobutane
(b) 1-Chlorobutane
(c) 2-Chloro-2-methylpropane
(d) 1-Chloro-2-methylpropane

Answer: (d) 1-Chloro-2-methylpropane — Quick Check 21.1.

21.1 PREPARATION OF HALOGENOALKANES

17. Ethane reacts with chlorine in the presence of light to give:

(a) Ethene
(b) Chloroethene
(c) Ethanol
(d) Chloroethane

Answer: (d) Chloroethane — Halogenation of alkanes.

18. Ethene reacts with HBr to give:

(a) Dibromoethane
(b) Bromoethane
(c) Ethanol
(d) Bromoethene

Answer: (b) Bromoethane — Electrophilic addition.

19. The catalyst in the reaction of an alcohol with HCl is:

(a) $ZnCl_2$
(b) $AlCl_3$
(c) Pyridine
(d) Ni

Answer: (a) $ZnCl_2$ — Method iii.

20. The catalyst used with thionyl chloride is:

(a) $ZnCl_2$
(b) $FeCl_3$
(c) Pyridine
(d) H_2SO_4

Answer: (c) Pyridine — Method iv.

21. The by-products of the thionyl chloride method are:

(a) H_2O and HCl
(b) H_3PO_3
(c) SO_3 and H_2
(d) SO_2 and HCl

Answer: (d) SO_2 and HCl — Both are gases.

22. The thionyl chloride method is especially useful because:

(a) The by-products are gases
(b) It is the cheapest
(c) It needs no alcohol
(d) It gives bromoalkanes

Answer: (a) The by-products are gases — The pure product is left behind.

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23. Ethanol reacts with PBr_3 to give bromoethane and:

(a) H_3PO_4 (b) H_3PO_3
(c) $POBr_3$ (d) HBr

Answer: (b) H_3PO_3 — Method v.

24. Ethanol reacts with PCl_5 to give chloroethane, HCl and:

(a) H_3PO_3 (b) PCl_3
(c) $POCl_3$ (d) H_3PO_4

Answer: (c) $POCl_3$ — Method v.

25. How many moles of ethanol react with one mole of PBr_3?

(a) One (b) Five
(c) Two (d) Three

Answer: (d) Three — From the balanced equation.

26. In the KBr/H_2SO_4 method, the acid first reacts with KBr to give:

(a) HBr
(b) Br_2
(c) SO_2
(d) K_2SO_4 only

Answer: (a) HBr — HBr then reacts with the alcohol.

27. Ethanol + KBr + H_2SO_4 gives bromoethane, water and:

(a) K_2SO_4 (b) $KHSO_4$
(c) KOH (d) SO_2

Answer: (b) $KHSO_4$ — Method vi.

28. Phosphoric acid is preferred for making iodoalkanes because it is a:

(a) Strong acid (b) Strong oxidizing agent
(c) Weak oxidizing agent (d) Base

Answer: (c) Weak oxidizing agent — It does not oxidize iodide to iodine.

29. Bromoethane is naturally emitted mainly by:

(a) Marine algae and phytoplankton (b) Volcanoes
(c) Forest fires (d) Cattle

Answer: (a) Marine algae and phytoplankton — Such as Trichodesmium.

30. In the atmosphere bromoethane plays a role in:

(a) Acid rain (b) Smog formation
(c) Stratospheric ozone depletion (d) Global cooling

Answer: (c) Stratospheric ozone depletion — Interesting Information box.

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21.2 REACTIVITY OF HALOGENOALKANES

31. The main factor deciding the reactivity of halogenoalkanes is:

(a) Strength of the carbon–halogen bond
(b) Bond polarity
(c) Molecular mass
(d) Colour

Answer: (a) Strength of the carbon–halogen bond — A weaker bond breaks more easily.

32. The bond energy of the C–F bond is:

(a) 346 kJ/mol (b) 290 kJ/mol
(c) 467 kJ/mol (d) 228 kJ/mol

Answer: (c) 467 kJ/mol — Table 21.1.

33. The bond energy of the C–Cl bond is:

(a) 467 kJ/mol (b) 346 kJ/mol
(c) 290 kJ/mol (d) 228 kJ/mol

Answer: (b) 346 kJ/mol — Table 21.1.

34. The bond energy of the C–Br bond is:

(a) 346 kJ/mol (b) 467 kJ/mol
(c) 228 kJ/mol (d) 290 kJ/mol

Answer: (d) 290 kJ/mol — Table 21.1.

35. The bond energy of the C–I bond is:

(a) 228 kJ/mol (b) 290 kJ/mol
(c) 346 kJ/mol (d) 467 kJ/mol

Answer: (a) 228 kJ/mol — The weakest C–X bond.

36. The most reactive halogenoalkanes are:

(a) Fluoroalkanes (b) Chloroalkanes
(c) Iodoalkanes (d) Bromoalkanes

Answer: (c) Iodoalkanes — Weakest bond.

37. The least reactive halogenoalkanes are:

(a) Iodoalkanes (b) Chloroalkanes
(c) Bromoalkanes (d) Fluoroalkanes

Answer: (d) Fluoroalkanes — Strongest bond.

38. In R–X the carbon attached to the halogen carries a:

(a) Partial negative charge (b) Partial positive charge
(c) Full negative charge (d) No charge

Answer: (b) Partial positive charge — The halogen is more electronegative.

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39. Which is NOT listed as a factor affecting reactivity of halogenoalkanes?

(a) Colour of the compound (b) Steric hindrance
(c) Inductive effect (d) Bond polarity

Answer: (a) Colour of the compound — Section 21.2.

21.3 REACTIONS OF HALOGENOALKANES

40. The carbon of the C–X bond acts as:

(a) A nucleophilic centre (b) A base
(c) A free radical (d) An electrophilic centre

Answer: (d) An electrophilic centre — It can be attacked by a nucleophile.

41. In nucleophilic substitution, the halide ion that departs is the:

(a) Substrate (b) Leaving group
(c) Nucleophile (d) Electrophile

Answer: (b) Leaving group — Section 21.3.

42. Removal of HX from a halogenoalkane gives an:

(a) Alkene (b) Alkane
(c) Alcohol (d) Amine

Answer: (a) Alkene — Elimination reaction.

43. The S_N2 mechanism takes place in:

(a) Two steps (b) Three steps
(c) One step (d) Four steps

Answer: (c) One step — Bond making and breaking occur together.

44. In S_N2 the nucleophile attacks from the side:

(a) Opposite to the leaving group (b) Same as the leaving group
(c) Above the molecule only (d) Of the hydrogen atoms

Answer: (a) Opposite to the leaving group — Back-side attack.

45. The order of an S_N2 reaction is:

(a) First (b) Zero
(c) Second (d) Third

Answer: (c) Second — Rate = k[RX][Nu].

46. The molecularity of an S_N2 reaction is:

(a) One (b) Zero
(c) Three (d) Two

Answer: (d) Two — Two molecules in the rate-determining step.

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47. S_N2 is always followed by:

(a) Tertiary halogenoalkanes (b) Primary halogenoalkanes
(c) Halogenoarenes (d) Alkenes

Answer: (b) Primary halogenoalkanes — Steric reasons.

48. The S_N1 mechanism takes place in:

(a) One step (b) No definite steps
(c) Three steps (d) Two steps

Answer: (d) Two steps — Ionization, then attack.

49. The intermediate in the S_N1 mechanism is a:

(a) Carbanion (b) Free radical
(c) Carbocation (d) Transition state only

Answer: (c) Carbocation — Formed in the first step.

50. The rate-determining step of S_N1 is:

(a) Attack of the nucleophile (b) Ionization of the alkyl halide
(c) Loss of a proton (d) Formation of the product

Answer: (b) Ionization of the alkyl halide — A covalent bond is broken; slow.

51. The rate of an S_N1 reaction depends on the concentration of:

(a) The alkyl halide only (b) The nucleophile only
(c) Both (d) The solvent only

Answer: (a) The alkyl halide only — Rate = k[Alkyl halide].

52. S_N1 reactions show:

(a) Second order kinetics (b) Zero order kinetics
(c) First order kinetics (d) Third order kinetics

Answer: (c) First order kinetics — Unimolecular.

53. S_N1 is always followed by:

(a) Tertiary halogenoalkanes (b) Primary halogenoalkanes
(c) Methyl halides (d) Halogenoarenes

Answer: (a) Tertiary halogenoalkanes — Stable tertiary carbocation.

54. The solvent mentioned for S_N1 ionization is:

(a) Benzene (b) Carbon tetrachloride
(c) Hexane (d) Aqueous acetone

Answer: (d) Aqueous acetone — Or aqueous ethyl alcohol.

55. The most stable carbocation is:

(a) Methyl (b) Tertiary
(c) Primary (d) Secondary

Answer: (b) Tertiary — Greatest +I effect.

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56. The least stable carbocation is:

(a) Tertiary (b) Primary
(c) Secondary (d) Methyl

Answer: (d) Methyl — No alkyl groups.

57. Carbocation stability is explained by the:

(a) Negative inductive effect
(b) Hydrogen bonding
(c) Positive inductive effect of alkyl groups
(d) Resonance of halogens

Answer: (c) Positive inductive effect of alkyl groups — Section 21.3.1.

58. A secondary halogenoalkane follows:

(a) Only S_N1
(b) Both S_N2 and S_N1
(c) Only S_N2
(d) Neither

Answer: (b) Both S_N2 and S_N1 — Depending on the solvent.

59. In β-elimination, the hydrogen is removed from the:

(a) β-carbon (b) α-carbon
(c) γ-carbon (d) Halogen

Answer: (a) β-carbon — The carbon next to the one bearing X.

60. β-Elimination occurs when the halogenoalkane is attacked by a:

(a) Nucleophile at carbon (b) Electrophile
(c) Free radical (d) Base at the β-hydrogen

Answer: (d) Base at the β-hydrogen — Section 21.3.2.

61. Elimination reactions often compete with:

(a) Addition (b) Oxidation
(c) Substitution (d) Polymerisation

Answer: (c) Substitution — They take place simultaneously.

62. The E2 mechanism occurs in:

(a) Two steps (b) One step
(c) Three steps (d) Four steps

Answer: (b) One step — Base attack and loss of X together.

63. The order of an E2 reaction is:

(a) Second (b) First
(c) Zero (d) Third

Answer: (a) Second — Rate = k[Alkyl halide][Base].

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64. The first step of the E1 mechanism is:

(a) Attack of the base
(b) Loss of hydrogen
(c) Slow ionization to a carbocation
(d) Formation of the alkene

Answer: (c) Slow ionization to a carbocation — The same as in S_N1.

65. 2-Bromo-2-methylpropane undergoes E1 elimination to give:

(a) But-1-ene
(b) 2-Methylpropene
(c) Propene
(d) But-2-ene

Answer: (b) 2-Methylpropene — Section 21.3.2.

66. Primary halogenoalkanes generally undergo elimination by:

(a) E1
(b) Free radical
(c) S_N1
(d) E2

Answer: (d) E2 — Section 21.3.2.

67. Tertiary halogenoalkanes undergo elimination by:

(a) E1
(b) E2
(c) S_N2
(d) Addition

Answer: (a) E1 — Stable carbocation.

21.4 EXAMPLES OF REACTIONS OF HALOGENOALKANES

68. The hydroxide ion is:

(a) Only a nucleophile
(b) An electrophile
(c) Only a base
(d) Both a good nucleophile and a good base

Answer: (d) Both a good nucleophile and a good base — It gives substitution and elimination.

69. Bromoethane with aqueous NaOH gives mainly:

(a) Ethene
(b) Ethanol
(c) Ethane
(d) Ethanal

Answer: (b) Ethanol — Substitution.

70. Bromoethane with alcoholic NaOH gives mainly:

(a) Ethanol
(b) Ethane
(c) Ethene
(d) Ethoxyethane

Answer: (c) Ethene — Elimination.

71. Reflux conditions are used to avoid:

(a) Loss of volatile products
(b) Oxidation
(c) Formation of precipitate
(d) Explosion

Answer: (a) Loss of volatile products — Section 21.4.

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72. The cyanide ion is a:

(a) Weak nucleophile and strong base
(b) Weak nucleophile and weak base
(c) Strong electrophile
(d) Strong nucleophile but weak base

Answer: (d) Strong nucleophile but weak base — So it gives substitution.

73. Bromoethane reacts with ethanolic KCN to give:

(a) Ethanenitrile (b) Butanenitrile
(c) Propanenitrile (d) Ethanamine

Answer: (c) Propanenitrile — The chain gains one carbon.

74. Bromoethane reacts with ammonia to give first:

(a) Ethanamide (b) Ethanamine
(c) Ethanol (d) Ethanenitrile

Answer: (b) Ethanamine — And HBr.

75. Ethanamine reacts with bromoethane to give:

(a) N-Ethylethanamine (b) N,N-Diethylethanamine
(c) Ethanamide (d) Propanamine

Answer: (a) N-Ethylethanamine — A secondary amine.

76. With a large excess of bromoethane, ammonia finally gives a:

(a) Primary amine (b) Nitrile
(c) Quaternary ammonium salt (d) Amide

Answer: (c) Quaternary ammonium salt — Four ethyl groups on nitrogen.

77. Further substitution occurs with ammonia because the amine formed is also a:

(a) Nucleophile (b) Base only
(c) Leaving group (d) Electrophile

Answer: (a) Nucleophile — It competes with ammonia.

78. In the silver nitrate test, the halogenoalkane is hydrolysed by:

(a) Ethanol (b) Nitrate ions
(c) Silver ions (d) Water

Answer: (d) Water — Water acts as the nucleophile.

79. Ethanol is used in the silver nitrate test to:

(a) Act as a catalyst (b) Dissolve the halogenoalkane
(c) Precipitate silver (d) Oxidize the halide

Answer: (b) Dissolve the halogenoalkane — Section 21.4.

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80. The colour of the AgCl precipitate is:

(a) Cream yellow (b) Bright yellow
(c) White (d) Black

Answer: (c) White — Identifies a chloroalkane.

81. The colour of the AgBr precipitate is:

(a) White (b) Red
(c) Bright yellow (d) Cream yellow

Answer: (d) Cream yellow — Identifies a bromoalkane.

82. The colour of the AgI precipitate is:

(a) White (b) Bright yellow
(c) Cream yellow (d) Blue

Answer: (b) Bright yellow — Identifies an iodoalkane.

83. Which halogenoalkane gives a precipitate with silver nitrate most slowly?

(a) Chloroalkane (b) Iodoalkane
(c) Bromoalkane (d) All at the same rate

Answer: (a) Chloroalkane — Strongest of the three bonds.

84. C_2H_5Br reacts with CH_3COONa to give:

(a) Ethanoic acid (b) Ethoxyethane
(c) Ethanol (d) Ethyl ethanoate

Answer: (d) Ethyl ethanoate — Quick Check 21.3 b.

21.5 HALOGENOARENES

85. Chlorobenzene is prepared from benzene and chlorine using:

(a) UV light (b) Anhydrous $FeCl_3$
(c) Nickel (d) Pyridine

Answer: (b) Anhydrous $FeCl_3$ — Section 21.5.

86. The preparation of halogenoarenes from benzene is an:

(a) Electrophilic addition
(b) Nucleophilic substitution
(c) Electrophilic aromatic substitution
(d) Elimination

Answer: (c) Electrophilic aromatic substitution — At room temperature.

87. The halogenation of benzene is typically carried out at:

(a) 500 °C (b) 150 °C
(c) –78 °C (d) Room temperature

Answer: (d) Room temperature — Section 21.5.

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88. In chlorobenzene the lone pair of chlorine is:

(a) Delocalised into the ring (b) Lost
(c) Donated to hydrogen (d) Localised on carbon

Answer: (a) Delocalised into the ring — Section 21.5.1.

89. The C–Cl bond in chlorobenzene has:

(a) Ionic character (b) Triple-bond character
(c) Partial double-bond character (d) No polarity

Answer: (c) Partial double-bond character — So it is stronger.

90. Compared with halogenoalkanes, halogenoarenes are:

(a) Less reactive (b) More reactive
(c) Equally reactive (d) Explosive

Answer: (a) Less reactive — And synthetically less useful.

91. Chlorobenzene with NaOH under normal conditions gives:

(a) Phenol (b) No reaction
(c) Benzene (d) Cyclohexanol

Answer: (b) No reaction — The C–Cl bond is difficult to break.

92. DDT has been used as a:

(a) Fertilizer (b) Refrigerant
(c) Anesthetic (d) Pesticide

Answer: (d) Pesticide — For mosquito control.

93. The use of DDT has been restricted because of its:

(a) High cost (b) Bad smell
(c) Persistence in the environment (d) Low effectiveness

Answer: (c) Persistence in the environment — Interesting Information box.

94. Removal of DDT in some developing countries led to an increase in deaths from:

(a) Malaria (b) Cholera
(c) Typhoid (d) Dengue only

Answer: (a) Malaria — It is a cheap mosquito control agent.

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